ZACKARY LOEVSETH

SELECTED RESEARCH PORTFOLIO

AI-Assisted Mathematics | Reproducible Research | Human-AI Systems

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PROFILE Independent researcher and research-program director working across computational mathematics, exact verification, scientific reproducibility, and human-AI research systems. I define targets, design research and validation protocols, direct AI-assisted proof and computation workflows, adjudicate corrections, and prepare evidence-bound publication packages.

SELECTED PUBLICATION-READY PREPRINTS

Explicit Counterexample Families to Five Finite-Group Inequalities from Graffiti3

FINITE GROUP THEORY / EXACT COMPUTATION

Result: Constructs infinite counterexample families to five proposed inequalities: U_7(q) for three conjectures and dihedral families for two others; also records a bounded SmallGrp census through order 256.

Methods: Symbolic family proofs, GAP, C, independent exact checking, fault injection, presentation-level reconstruction, reproducibility replay.

Status: PUBLICATION-READY PREPRINT — EXTERNAL SPECIALIST REVIEW PENDING.

Largest-Index Zonotope Bounds for Planar Projections of Semiorder Ideals

COMBINATORIAL OPTIMIZATION / ORDER THEORY / GEOMETRY

Result: Proves instance-sensitive hull bounds h(P,v) <= 1+n+inc(P), f(P,v) <= 1+n+2inc(P), and O(nr) upper-hull complexity for width-r represented semiorders.

Methods: Combinatorial proof, planar zonotope geometry, exact rational enumeration, source-independent verifier, clean-archive zero-mutation replay.

Status: PUBLICATION-READY PREPRINT — EXTERNAL SPECIALIST REVIEW PENDING.

Excess-Degree Bounds for Minimal Counterexamples to the Erdos-Gyarfás Conjecture

GRAPH THEORY

Result: For a lexicographically minimal hypothetical counterexample, proves |A| >= 2b+s+4 and derives density and edge-count consequences; the global conjecture remains open.

Methods: Definition-level proof reconstruction, suppression-graph analysis, hostile audit, independent falsification tools, mutation testing, reproducibility archive.

Status: PUBLICATION-READY PREPRINT — EXTERNAL SPECIALIST REVIEW PENDING.

PORTFOLIO STATUS Three distinct mathematics manuscripts have completed internal proof, reproducibility, scope, citation, and publication-preparation gates. They are preprints, not peer-reviewed publications, and none has yet been submitted.

RESEARCH PROGRAMS AND METHODS

HUMAN-AI RESEARCH SYSTEMS

Theory of Epistemic Control / RR-ETI

Developed a decision-and-provenance framework for correction, dependency repair, authority, target preservation, and completion control in long-horizon human-AI work.

Study 14

Designed a living validation program and frozen four-condition confirmatory core for premise-governed context scheduling; comparative model execution has not yet begun.

Autonomous Scientific Director

Directed a portfolio-scale workflow that selected targets, corrected failed forecasts, ran independent checks, and moved three separate mathematics campaigns to publication-ready packages while reserving final public action for the human.

MATHEMATICAL RESEARCH PROGRAMS

Erdos Problem #409

Public exact certificates through F=71 and a recovered F=104 packet awaiting fresh replay and priority audit; global basin and density questions remain open.

Riemann Hypothesis

Directed a governed research program on rational-bump Weil reductions, computable test functions, conditional Li-witness bounds, and proof-carrying evaluator design. RH remains unsolved.

Parametric Closure

Publication-ready bounded-width semiorder theorem; unrestricted mixed-sign parametric closure remains open.

Applied GIS / Wildfire

Completed a substantial dynamic wildfire-risk and hazard-mapping thesis/case study; derivative publication and updated validation remain optional next steps.

TECHNICAL METHODS

Python, Ruby, C, shell scripting, Git; GAP and exact finite-group computation; exact rational and interval arithmetic; FLINT/Arb and python-flint; Lean/proof-assistant workflows; graph and combinatorial enumeration; SAT/certificate reasoning; producer-checker separation; hostile audit; mutation and corruption testing; deterministic builds; SHA-256 manifests; clean-checkout replay; scientific manuscript and reproducibility-package production.

ROLE AND AI DISCLOSURE

My role includes target selection, research-contract design, route and resource decisions, verification standards, correction adjudication, evidence preservation, and publication boundaries. AI systems substantially assisted with literature search, proof derivation, computation, independent checking, recovery, and drafting. All manuscripts carry explicit AI-use disclosures; external specialist review remains pending.

SELECTED EVIDENCE HIGHLIGHTS

3	318,667	343,980	F=71
publication-ready mathematics packages	exact semiorder cases replayed	admissible graph-theory tuples checked	certified public scope through F=71

TARGET ROLES

AI research and evaluation | Research engineering | Scientific computing | Reproducibility and verification | Research operations and program direction | Technical writing and evidence synthesis

PERMANENT PORTFOLIO: https://zackaryloevseth.github.io/research-portfolio/

Full manuscripts, code, exact verification, reproducibility archives, research-program briefs, and resume.